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Community Detection in Online Social Media Using Social Networks Analysis; A Comparative Study

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Abstract - Group of users that are intra densely connected with each other are termed as the community over an online social network. People over an online social network group together in these communities and most of the times they share the similar interests. In this project we are going to detect these communities over these online social networks using the techniques of Social Network Analysis (SNA) such as clique percolation method, random walks and Louvain methods. The proposed approach will be tested on real social network datasets and the experimentations will be done in python. The empirical venture will show the best results and can have best implications in real world situations.

Keywords – Clique percolation method, Community Detection, Influential user, Louvain methods, Online Social Networks, random walks Social Graphs, Social Network Analysis

1. Introduction

The growth of social networks has changed the way people interact and share information and this has created a huge amount of user data [1]. In these networks people often form groups where they have a lot of connections with each other. Not as many with people outside of the group. Find these groups is important when we study networks [2]. This is because it helps us understand how people are connected to each other how information spreads from person to person and how social networks groups actually work [3]. We need to know about social networks groups to see how they interact with each other and with people, outside of the social network group [4]. Understanding these groups is really important for things like building systems that recommend things to people creating targeted marketing plans and stopping information from spreading [5].

Old ways of finding these groups do not work well because social networks are complicated and always changing and people often belong to many groups at the same time [6]. Hence, it becomes necessary to use techniques that can find overlapping communities [7]. Nowadays, the size of social networks is increasing so much that algorithms have to be created that could process a large amount of information [1]. In this research, certain popular algorithms used for the study of social networks such as Louvain, clique percolation, and random walk methods would be reviewed and tested in relation to their ability to detect communities in social media. It would be studied through the application of these methods on data and with the help of the programming language Python whether the methods under review are effective, feasible, and significant [8]. Social Network Analysis, also referred to as SNA, would be used for this purpose [9]. Online Social Networks (OSNs) would be the subject of our study.

2. Related Work

A diverse array of algorithms has been developed for community detection, broadly categorized into methods based on graph partitioning, hierarchical clustering, optimization, and spectral analysis. Recent advancements also encompass deep learning and graph neural network (GNN) approaches, which offer enhanced capabilities for handling complex network structures and attributed social networks [11][12]. For instance, DEGCN employs graph embedding within a deep learning framework to improve accuracy and speed in attributed social networks. Furthermore, community detection has found applications beyond traditional social networks, extending to manufacturing communities and scientific literature networks [4].

Modularity as a Measure Modularity is a widely accepted metric to quantify the strength of a community division within a network [10][4]. It evaluates the difference between the actual number of edges observed within communities and the expected number of edges in a random network with an identical number of nodes and edges but random connections. A higher modularity value indicates a more pronounced and robust community structure. The goal of community detection algorithms is to make the groups of nodes in a network separate as possible. This is often called optimizing modularity. Lots of people try to do this. It is a really tough problem to solve. We have to use kinds of algorithms to get an answer.

The formula to calculate modularity, which we call Q is given by:

Complete Formula of Modularity (Q)

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(g_i, g_j)$$

- A_{ij}→ 1 if there is an edge between node i and node j, otherwise 0
- k_i→ Degree of node i (number of edges connected to i)
- k_j→ Degree of node j
- 2m→ Total number of edges in the network × 2
- $\delta(g_i, g_j)$→
  - 1 if nodes i and j are in the same community
  - 0 if they are in different communities

We use modularity to see how well the nodes in a network are grouped together. Optimizing modularity helps us find the groups of nodes, in a network.

3. Literature Survey

Community detection in social media is really important for understanding how people are connectedFinding groups of people who're more connected to each other than to others is really important [1]. This is because it helps us understand how people talk to each other and how information moves around [10]. It is a deal because groups of people who are more connected to each other than to people, outside their group help us understand how people interact with each other and how information spreads [13]. We can learn a lot from these groups of people.

It also has a lot of uses like helping to stop the spread of false information and making advertising more targeted.

People have come up with ways to study these groups and each way has its good and bad points. Community detection in social media is essential for understanding how people are connected but it is still not easy to do even with new and better algorithms like the Louvain method the Clique Percolation Method and methods that use random walks.

Community discovery is essential for studying people's social connections; however, it is still difficult despite recent algorithm advancements, such as the Louvain method, Clique Percolation Method (CPM), and random walk algorithms. While the Louvain method is effective in large networks, it has limitations in terms of the "resolution limit" and sensitivity to the order of nodes processing that can result in the failure to identify smaller communities. CPM can accurately find overlapping communities, recognizing multiple group memberships of users [14].

The concept of random walk involves finding hidden patterns of social relationships by simulating movement across dense parts of a network [10]. As social networks continue evolving, current methods mainly focus on the analysis of their structural properties. Consequently, these approaches ignore temporal changes in dynamics and semantic meaning of interactions. Hence, they cannot always give accurate results.

There are no tools available to assess the performance of these methods. Social networks evolve rapidly. Therefore, it is challenging to measure effectiveness. Moreover, any static approach cannot reveal community dynamics properly [15].

In order for us to improve our techniques, we have to utilize real-time data, which will include the actions and emotions of the user [16]. In addition, it would be necessary for us to integrate the techniques with emerging ones, such as deep learning and graph neural network approaches [6].This will help us find communities in an informed way. Social networks and community identification need this to be more accurate. Community identification will be better, with these changes [3][17].

4. Methodology

This study looks at three ways to find communities: the Louvain method, the clique percolation method and methods that use random walks. The Louvain method, the clique percolation method and methods that use walks were chosen because they are different and have their own way of finding communities. The Louvain method, the clique percolation method and methods that use walks are good, at handling big networks and finding communities that overlap with each other.

The random walk algorithm for community detection is based on the idea that when a random walker is moving in a network, it has a higher chance of being in a densely linked community for a longer period of time before proceeding to another region of the network [9]. For example, influential nodes, which are usually the central points of communities, can be detected using the eigenvector centrality method, and users can be assigned to communities according to their structural similarity and behaviour obtained through random walk algorithms [6]. This is an efficient way of detecting hidden group behaviours in a network [9].

The Clique Percolation Method (CPM) is a method that is particularly tailored to detect overlapping communities, recognizing that people are often members of several social groups at the same time. It describes communities in terms of the notion of k-cliques, which are maximal subgraphs in which every k-nodes are connected [14]. The approach consists of finding all k-cliques in a graph, building a "clique graph" in which vertices are the k-cliques, and finally, defining communities as the set of vertices in the connected components of the clique graph. The CPM is especially useful in detecting vertices that serve as bridges between different communities [14].

The Louvain algorithm is a fast method that many people use to find groups in big networks [10]. It works by trying to make these groups as good as possible. The algorithm does this in two steps. First it tries to make the groups better by moving one node at a time to a group. Then it combines some groups into ones to make a simpler network [13]. The Louvain algorithm keeps doing this until it cannot make the groups better [18].

The goal is to make the groups, called communities as strong as possible which is measured by something called modularity. The Louvain algorithm is good at finding communities in graphs by maximizing modularity. It repeats the process of moving nodes and combining groups until no further improvement, in modularity is possible. the Louvain algorithm is based on this iterative two-step process.

Despite being computationally efficient and scalable for big data analytics, there is a downside to this algorithm as it is prone to having "resolution limit," where it fails to detect small communities. It can also be sensitive to the node ordering process [13]C. There are modifications to the algorithm, for example, H-Louvain to deal with these challenges, especially when analyzing dynamic streams of social media data.

Algorithm 1:

Input: Social Network Graph G (V, E)
Output: Communities Com1, Com2, Com3
1. Detect Communities using CPM
   Uncover k-cliques
   Detect communities → Com1
2. Community Detection using Random Walk Algorithm
   Perform random walk process on graph G
   Cluster nodes → Com2
3. Community Detection using Louvain Method
   Optimize modularity
   Form communities → Com3
4. **Return** Com1, Com2, Com3

The above methods will be tested on different social network data sets to ensure their generalizability. Some suitable datasets to test these algorithms are:

1. **Snap Social Circles Dataset** – a dataset developed from the Facebook dataset which gives a ground truth community and allows comparative analysis with known communities [19][20]
2. **Twitter Network Datasets** – A topic-based network dataset instead of the more popular follower-based networks which give rise to community formation by similar interests and ideas [1].
3. **Large Scale Diverse Graphs:** Diverse large scale graphs can also be used to evaluate the performance of the proposed algorithm in terms of its applicability to diverse large scale graphs, which have dynamic nature in many cases. Such graphs may include the Amazon products graph, Bitcoin Transaction Graph, and Cellular Phone Network graphs [7], [20].

5. Results and Discussion

The findings reveal three major clusters, thus implying that random walkers have a tendency to move within well-connected zones of the graph. This method effectively captures structural similarity among users. The Proposed approach will be tested on real social network datasets and the experimentations will be done in python. The empirical venture will show the best results and can have best implications in real world situations.

Finally, the integration of semantic information and natural language processing with community detection algorithms could result in more meaningful and context-aware community detection in platforms that are rich in textual data.

Community detection in social media is really interesting. It involves looking at the ways to find groups of people who interact with each other online. We can do this by comparing the algorithms that are out there. community detection in social media is important because it helps us understand how people behave online. we can see how these algorithms work in life and what they can tell us about online social interactions and community detection in online social media.

This can give us an idea of what is going on in online social media and community detection, in online social media. this work also aims to lay the foundation for future developments. The below are the graphs for a dataset using methods like random walks [9], clique percolation method [14] and Louvain method [10].

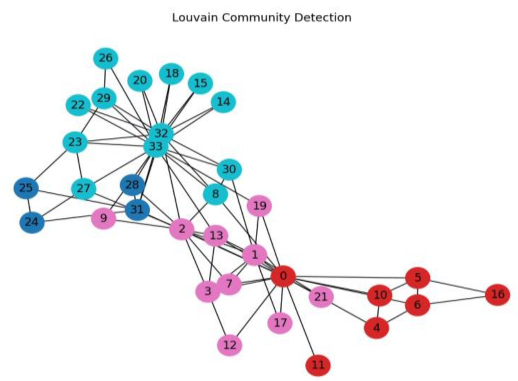

Fig.1, "The resulting graph, generated by the Louvain algorithm, displays a clear community structure with dense intra-group connections and sparse connections."

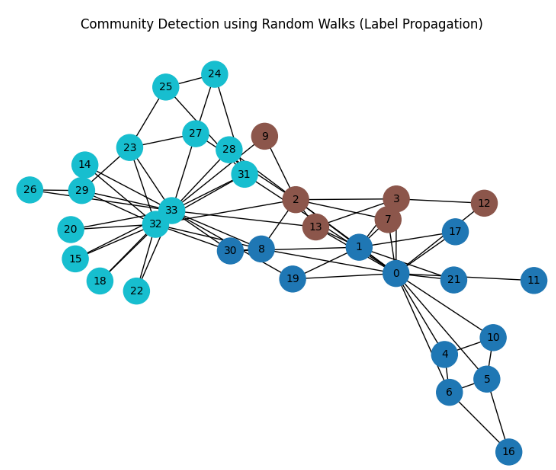

Fig.2, "The figure displays the paths generated by random walks on a dataset represented as a graph, illustrating the exploration of node connectivity and the distribution of visits across different nodes"

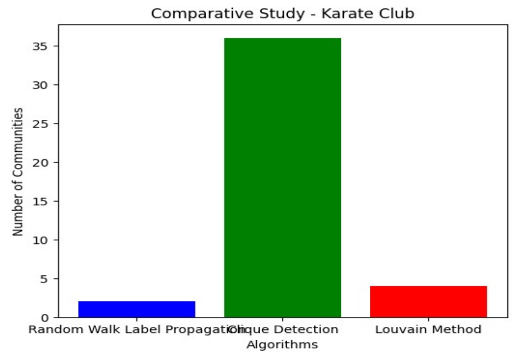

***Fig.3:** Comparative student on the Karate Club Network Dataset*

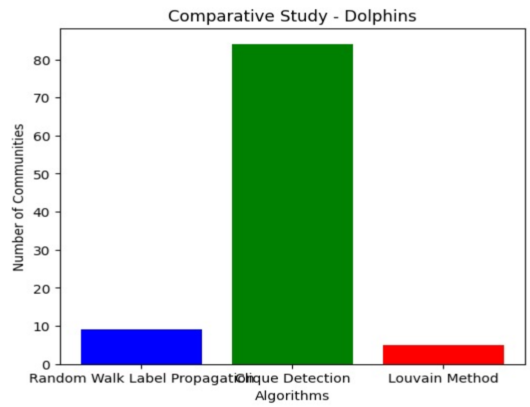

***Fig. 4**: Comparative student on the Dolphin Network* Dataset

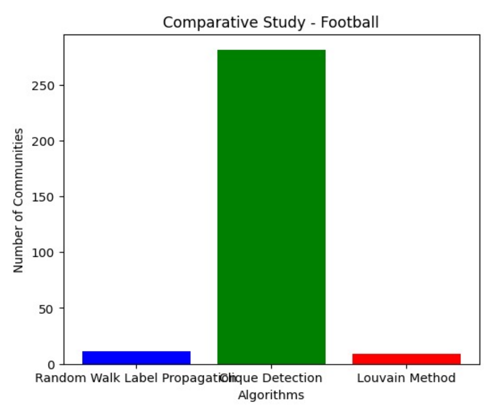

***Fig. 5:** Comparative student on the Dolphin Network Dataset*

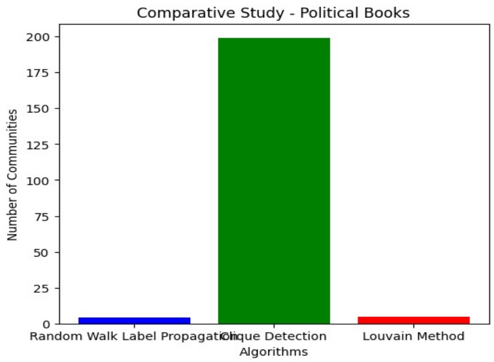

Fig. 6: *Comparative student on the Political Books Network Dataset*

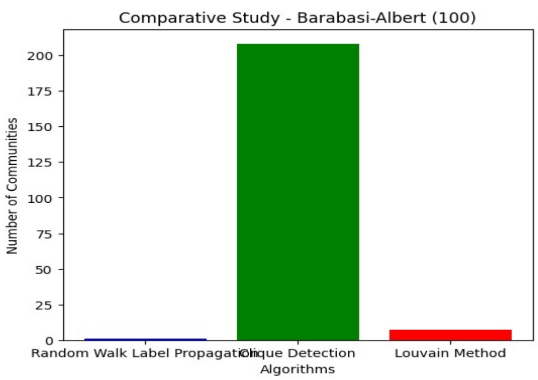

Fig.7: *Comparative study on the Barabasi Network Dataset*

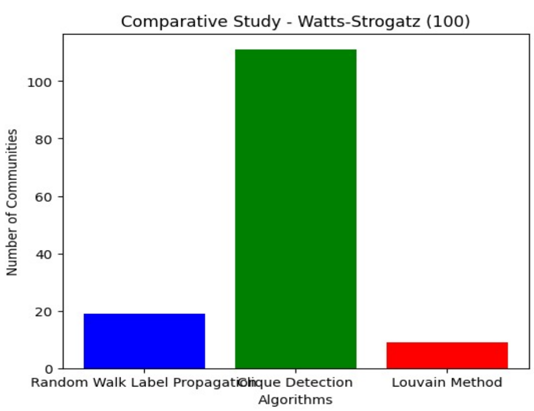

Fig.8: *Comparative study on the Watts stogatz Network Dataset*

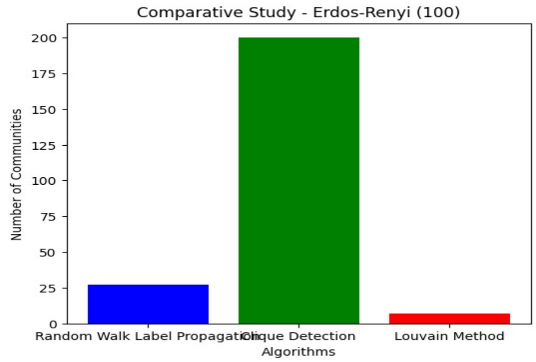

Fig .9: *Comparative study on the Erdos Renyl Network Dataset*

On experimentations we have got the following comparative results of all the methods over each dataset as shown in figures Fig. 3 through Fig. 9 below is the overall implementation of comparative study in graph of all the algorithms.

Despite the remarkable development achieved in the field of clustering, there are still some limitations associated with this process [13]. Most of the conventional techniques for discovering clusters rely solely on the structure of the graph, disregarding such aspects of nodes as their characteristics, which may affect the quality of clustering negatively [15]. Additionally, there is an issue related to the dynamics of the system as new clusters emerge and existing ones disintegrate constantly.

Moreover, the resolution limitation of algorithms such as Louvain will be a factor that hinders the discovery of smaller but still relevant communities [11]. Also, the subjective aspect of what qualifies as a "community" and the lack of standardized ground-truth data sets for most real-world networks complicate objective assessments [17]. Overlapping community detection, although essential, poses challenges in the definition and assessment of community borders [21].

6. CONCLUSION AND FUTURE WORK

In this study, an empirical comparison of some of the critical community detection techniques, including the Louvain algorithm, clique percolation algorithm, and random walk algorithms, on online social networks was performed. The findings were extremely valuable in the more profound understanding of community detection in online social networks. In future we will scale this study to the broader contours of dynamic evolving online social networks.

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